4. This problem examines possible biochemical explanations for variations of Mendel's 9:3:3:1 ratio. Except where indicated, compounds 1, 2, 3 and 4 have different colors, as do mixtures of these compounds. A and B are enzymes that catalyze the indicated steps of the pathway. Alleles *A* and *B* encode functional enzymes A and B, respectively; these are completely dominant to alleles a and b, which do not encode any of the corresponding enzyme. If functional enzyme is present, assume that the compound to the left of the arrow is converted completely to the compound to the right of the arrow. For each pathway, what phenotypic rations would you expect among the progeny of a dihybrid cross of the form *AaBb* × *AaBb*?

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a. Independent pathways

Compound 1 $\xrightarrow{\text{Enz A}}$ Compound 2

Compound 3 $\xrightarrow{\text{Enz B}}$ Compound 4

b. Sequential pathway

Compound 1 $\xrightarrow{\text{Enz A}}$ Compound 2 $\xrightarrow{\text{Enz B}}$ Compound 3

c. Redundant pathways

Compound 1 $\underset{\text{Enz B}}{\overset{\text{Enz A}}{\rightrightarrows}}$ Compound 2

AaBb × AaBb

↓

9 A_B_ : 3 A_bb : 3 aaB_ : 1 aabb

↓ ↓ ↓ ↓

p_1 p_2 p_3 p_4 (phenotype)

a. $p_1 : C_2 + C_4$ $p_2 : C_2 + C_3$ $p_3 : C_1 + C_4$ $p_4 : C_1 + C_3$.

$p_1 : p_2 : p_3 : p_4 = 9 : 3 : 3 : 1$

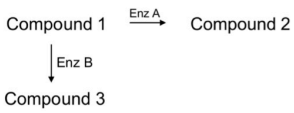
b. $p_1 : C_3$ $p_2 : C_2$ $p_3 : C_1$ $p_4 : C_1$

$p_1 : p_2 : (p_3 + p_4) = 9 : 3 : 4$

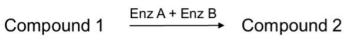
c. $p_1 : C_2$ $p_2 : C_2$ $p_3 : C_2$ $p_4 : C_1$

$(p_1 + p_2 + p_3) : p_4 = 15 : 1$

d. Branched pathways (assume a limitless supply of compound 1)



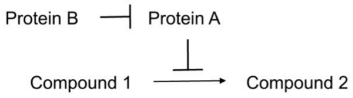
e. Complementary gene action (enzymes A and B both needed to catalyze the reaction indicated)



f. Now consider independent pathways as in (a), but the presence of compound 2 masks the colors due to all other compounds.

g. Next consider the sequential pathway shown in (b), but compounds 1 and 2 are the same color.

h. Finally, examine the pathway shown below. Here, compounds 1 and 2 have different colors. Protein A prevents the conversion of compound 1 to compound 2. Protein B prevents protein A from functioning.



d. $P_1 : (C_2 + C_3)$ $P_2 : C_2$ $P_3 : C_3$ $P_4 : C_1$

$P_1 : P_2 : P_3 : P_4 = 9 : 3 : 3 : 1$

e. $P_1 : C_2$ $P_2 : C_1$ $P_3 : C_1$ $P_4 : C_1$

$P_1 : (P_2 + P_3 + P_4) = 9 : 7$

f. $C_1 \xrightarrow{EA} C_2$

$C_3 \xrightarrow{EB} C_4$

$P_1 : C_2$ $P_2 : C_2$ $P_3 : C_1 + C_4$ $P_4 : C_1 + C_3$

$(P_1 + P_2) : P_3 : P_4 = 12 : 3 : 1$

g. $C_1 \xrightarrow{EA} C_2 \xrightarrow{EB} C_3$

$P_1 : C_3$ $P_2 : C_2$ $P_3 : C_1$ $P_4 : C_1$

$P_1 : (P_2 + P_3 + P_4) = 9 : 7$

h. $P_1 : C_2$ $P_2 : C_1$ $P_3 : C_2$ $P_4 : C_2$

$(P_1 + P_3 + P_4) : P_2 = 13 : 3$

1. What are the advantages and the limitations of non-invasive prenatal testing (NIPT)?

1. Advantages:

①NIPT 避免了羊水穿刺等侵入式检测方法对母体和胎儿的伤害。

②NIPT对 common autosomal trisomies 的检测具有高敏感性和高特异性，假阳性0.03%.

③孕八周即可检测

Limitations:

①NIPT研究的是胎盘而不是胎儿的DNA

②NIPT的失败率在2~6%

③21三体只有约50%的胎儿有染色体有畸变.

2. Assume parents with AA and BB alleles for mother and father respectively in the 18th chromosome and there are about 12% of fetal DNA detected in the NIPT. If FSR_{18}/FSR_{ref}= 0.94, what is the likely genotype of the baby? How can you be sure?

normal genotype of the baby is AB.

Ma Fetal

A A A B

$\frac{1-f}{2}$ $\frac{1-f}{2}$ $\frac{f}{2}$ $\frac{f}{2}$

$FSR_{ref} = \frac{\frac{f}{2}}{1-\frac{f+f}{2}+\frac{f}{2}} = \frac{f}{2-f}$

由于 $\frac{FSR_{18}}{FSR_{ref}} = 0.94 < 1$,

所以假设 the genotype of the baby is AAB

Ma Fetal

A A A A B

$\frac{1-f}{2}$ $\frac{1-f}{2}$ $\frac{f}{2}$ $\frac{f}{2}$ $\frac{f}{2}$

$FSR_{18} = \frac{\frac{f}{2}}{1-\frac{f+f}{2}+\frac{f}{2}+\frac{f}{2}} = \frac{f}{2}$

$\frac{FSR_{18}}{FSR_{ref}} = \frac{\frac{f}{2}}{\frac{f}{2-f}} = 1-\frac{f}{2}$

$f=12\%$, $1-\frac{f}{2} = 1-0.06 = 0.94$ 与题目相符

∴假设成立

The likely genotype of the baby is AAB.